

# Prince Yadav

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## PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building end-to-end AI systems in healthcare, finance, and nutrition domains. Proficient in C++, Python, SQL, deep learning, and LLMs. Focused on scalable, data-driven, and interpretable AI solutions with a track record of integrating ML models into production-ready web applications.

## RELEVANT COURSEWORK

Data Structures & Algorithms · Machine Learning & Deep Learning · LLMs & Generative AI · Computer Vision · Explainable AI & RAG · Data Mining · Predictive Analytics & Data Visualization · Software Development Life Cycle (SDLC)

## EXPERIENCE | INTERNSHIP

### AI / Machine Learning Intern · IBM

Jun 2025 – Jul 2025

#### *ExpenseXchange – AI-Powered Expense & Stock Portfolio Analyzer*

- Built an AI-based stock portfolio analyzer predicting short-term trends with Buy / Hold / Sell signals.
- Trained GRU-based time-series models on 5+ years of historical stock data across 20+ tickers, achieving ~76% directional accuracy and ~0.82 R<sup>2</sup> score.
- Reduced prediction error by ~7–10% using Huber Loss and Optuna hyperparameter tuning.

### Founder · NextAQLabs – Industrial AI Automation (Startup)

Jan 2024 – Present

- Designed AI-powered industrial automation systems that improved operational efficiency by 40–60%.
- Built smart, scalable AI solutions for real-world automation problems, achieving 2x–3x throughput gains.
- Developed hands-on AI and robotics education programs preparing students for future tech careers.

## PROJECTS

### Neolytics – Deep Learning-Based Cancer Detection with Explainable AI

Jan 2025 – Jun 2025

- Built an AI-powered cancer detection system that analyzed 1,000+ medical images to support early and accurate diagnosis.
- Implemented CNN-based transfer learning models achieving ~72% classification accuracy with Grad-CAM for model explainability.

### NutriCareHub – Personalized Diet Recommendation System

Aug 2024 – Dec 2024

- Designed an ML system predicting optimal diets based on user parameters including BMI, symptoms, allergies, and hereditary conditions.
- Built symptom-to-disease mapping search functionality with corresponding diet plan display.

### Echo-D – AI-Powered Ultrasound Preprocessing & Diagnostic Support

Aug 2025 – Dec 2025

- Developed an AI-based system for enhanced ultrasound analysis using SLIC, SAM, BLIP-2, and Grad-CAM++ techniques.
- Applied CLAHE and Otsu preprocessing pipelines to improve image clarity and diagnostic support accuracy.

## TECHNICAL & SOFT SKILLS

**Languages:** C++, Python, SQL, React

**Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Hugging Face Transformers

**AI / ML:** LLMs, RAG, Transfer Learning, Computer Vision, Explainable AI, GRU/LSTM

**Dev Tools:** Git & GitHub, VS Code, Google Collab, Anaconda, Local GPU/CPU

**Soft Skills:** Problem-solving, Team Collaboration, Public Speaking, Event Organizing

## EDUCATION

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**B.Tech in Computer Science – University of Petroleum and Energy Studies (UPES)**

Aug 2022 – Present

CGPA: 7.2 / 10

## CERTIFICATIONS & EXTRACURRICULARS

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- IBM Certification: Advanced Machine Learning Techniques · 2nd Runner-Up at IBM
- UPES DSW Ambassador – Facilitated student-administration communication and organized campus Events Jan 2024 – Feb 2025